Activity 2

Take one or two drops of hay (In tamil, vaikol) decoction on a slide and observe it under the microscope.

16.5.1 Cell structure of Protozoa

Amoeba is a unicellular microscopic organism. It is found in ponds. *Amoeba* is irregular in shape. It has cell membrane, cytoplasm and nucleus. It is a protozoan that moves by means of pseudopodia (In Latin, 'false feet'). Pseudopodia are the extended part of cell membrane. It helps to catch its prey (Algae). The body flows around the food particle and engulfs it forming food vacuoles. Contractile vacuoles are seen in the cytoplasm that help in excretion. *Amoeba* reproduces by means of fission and sporulation.

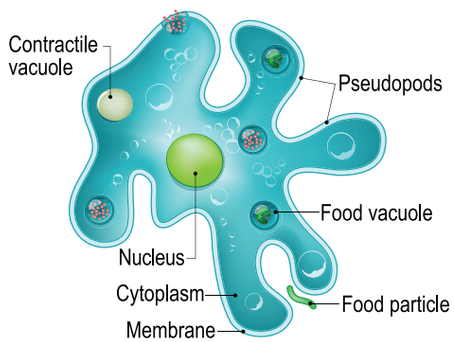


Figure 16.9 Amoeba

16.6 Prions

The word prion is derived from 'protinaceous infectious particle'. Prions have neither DNA nor RNA to transmit infection. A prion is a mutted form of a usually harmless protein. Prions cause diseases by affecting brain or neural tissue. Eg. Creutzfeldt-Jackob disease. Another example is Kuru- associated with cannibalism.

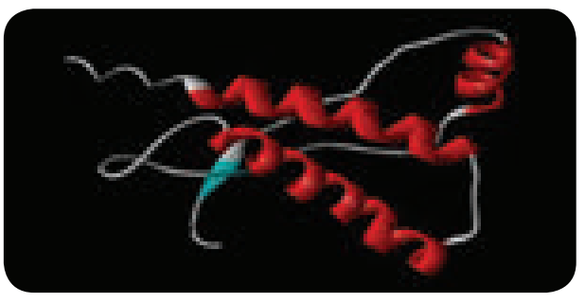


Figure 16.10 Structure of Prion

16.7 Virions

Virion is an entire virus particle consisting of an outer protein shell called a capsid and an inner core of nucleic acid (RNA or DNA). If the virus is found outside the cell (extracellular) it is known as virion. Virion has the capacity to infect the living tissue.

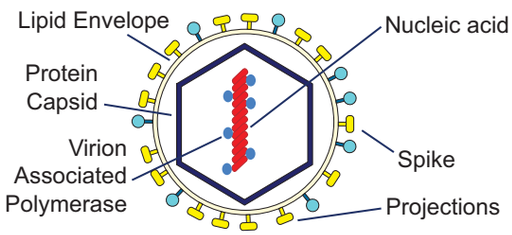


Figure 16.11 Structure of Virion

16.8 Uses of Microorganisms

Microorganisms are useful in different fields such as medicine, agriculture and industry. Some of them are given below.

16.8.1 Medicine

We obtain antibiotics and vaccines from microbes.

Antibiotics

The word 'anti' means 'against'. Antibiotic is a substance produced by living organisms which is toxic for other organisms.



Figure 16.12 Penicillium chrysogenum

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